

Architecture — FieldFlow Connector Prototype

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Deliverable: 01_Architecture.pdf

Runtime: .NET 10 modular monolith + local FieldFlow mock

Delivery model: at-least-once + idempotent effects (never exactly-once)

1. Executive summary and key decisions

This prototype proves a reusable Proof360 connector pattern against a local FieldFlow mock: durable inbound sync (poll + webhook), canonical Vendor/Job protection, qualified outbound dispatch via outbox, bounded HTTP resilience, and operator-visible health/audit.

Decision	Choice	Why
Shape	Modular monolith + hexagonal ports	Explainable Phase 1; architecture tests enforce boundaries
Delivery	At-least-once + idempotency	Exactly-once across HTTP/DB is not honest
Canonical boundary	Only Vendor / Job / Transcript	Assignment lock; connector metadata is sidecar
Persistence	SQLite now → PostgreSQL later	Local demo without infra drag (ADR-004)
TX vs HTTP	Never hold DB TX across provider I/O	Prevents stuck locks and partial dual-writes (ADR-005)
Commercial next	Accounting first (discovery-gated)	Finance ROI + safer than payments-first (ADR-008)

2. Constraints and assumptions

Facts: only three canonical entities/fields; FieldFlow DTOs stop at the ACL; no live PRG credentials; mock shares no production DTO assembly.

Assumptions (replace with real FieldFlow OpenAPI): contractor/work-order JSON shapes, HMAC over {unix}. {rawBody}, status vocabulary, ±300s webhook skew, stable outbound idempotency key fieldflow:{instance}:{jobId}:dispatch:v1. Full list: docs/assignment/assumptions.md.

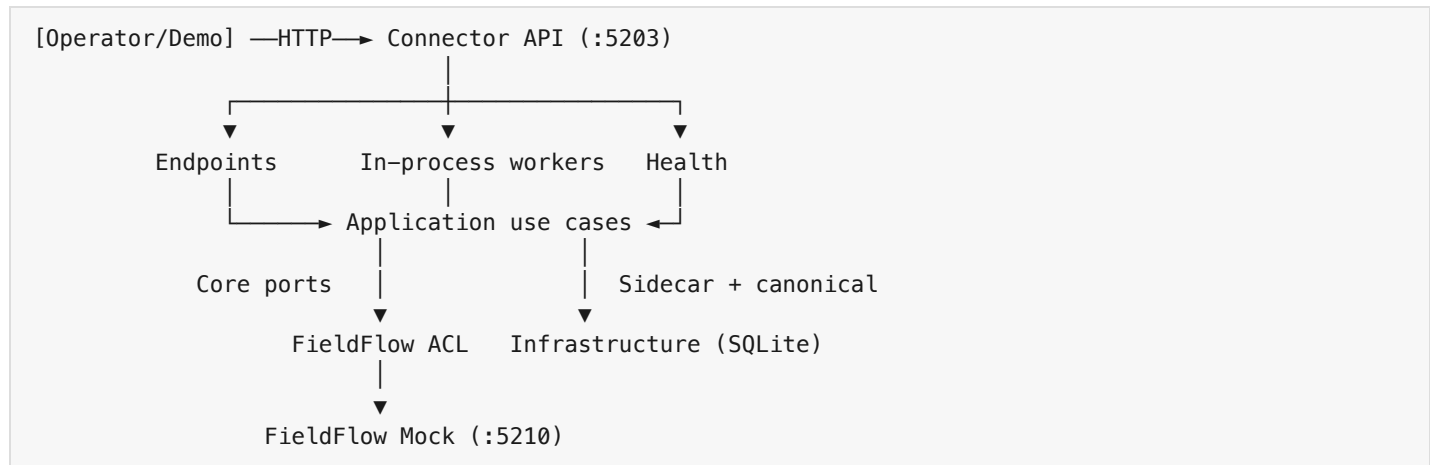
Approved deviation: .NET 10 (kit referenced .NET 8; only SDK 10 available here).

3. Phase 1 scope and exclusions

In scope: Contractor→Vendor and WorkOrder→Job import; poll+webhook inbox; HMAC; version ordering; unknown-contractor deferral; DLQ+replay; qualified Job outbox dispatch + ambiguous POST reconcile; retry/circuit/health; audit/correlation; four test projects.

Deferred: Invoice/Payment/Location/Appointment as canonical; automatic payments; multi-provider admin UI; production broker/OAuth/vault; full alert routing; Transcript sync; fuzzy matching. See docs/assignment/phase-1-scope.md.

4. Component and data flow



Projects: Domain (canonical + policies) → Core (ports/contracts/integration records) → Application (orchestration) → FieldFlow (ACL) / Infrastructure (EF) → API (composition). Mock has **zero** connector project references.

Diagram source: docs/architecture/component-and-data-flow.mmd.

5. Shared connector vs FieldFlow adapter

Shared (Application/Core)	FieldFlow-specific
Inbox/outbox/identity/audit models	JSON DTOs + HTTP client
Apply/disposition/eligibility policies	HMAC verifier + normalizer
Failure codes / Result types	Resilience pipeline wiring
Capability ports	Port implementations + mappers

A second provider adds a new adapter project implementing the same Core ports — not a fork of Application.

6. Canonical boundary vs connector metadata

Canonical types live only in Domain and use exact assignment column names.

ProviderIdentityLink, InboxMessage, OutboxMessage, AuditEvent, ConnectorState are **integration infrastructure**, not new Proof360 entities (ADR-002).

Never store FieldFlow IDs in vendor_id / job_id. Never dump provider JSON into ai_json / rationale / notes_scope. AI fields remain null unless an actual AI process produced them (Phase 1: always null).

7. Contractor → Vendor mapping

FieldFlow	Proof360	Ownership
contractorId	Identity link only	External identity
complianceId / license / insurance / WCB	Matching Vendor columns	FieldFlow supplies
expired / missing compliance	missing_items + status	Deterministic Proof360
active	May restrict	Cannot auto-approve
displayName	—	Ignored
—	created_at	Set once by Proof360
—	ai_confidence	Null

Approval asymmetry: first active import → pending_review; expired/restricted → restricted; reactivation alone cannot restore approved.

8. WorkOrder → Job mapping

FieldFlow	Proof360	Ownership
workOrderId	Identity link only	External identity
customer / address / service / notes / window	Job fields	Init on FieldFlow-origin; Proof360 owns on Proof360-origin
contractorId	assigned_vendor_id	Via identity resolve in Application
status	Job status	Map + monotonic transition policy
entityVersion	Sidecar / ordering	Not a canonical column
unknown optional JSON	Observability metadata	Never canonical storage

9. Field and status source of truth

Origin rule: Proof360-originated Jobs reject provider overwrite of customer/address/service/priority/window/notes/compliance_only. After confirmed dispatch, FieldFlow is authoritative for operational status progress under the transition policy.

FieldFlow status	Job status
open	dispatched
scheduled	scheduled
in_progress	in_progress
done	completed
void	cancelled

Also: Proof360 qualified before outbound. Terminal: completed / cancelled. Unknown statuses fail validation (no fuzzy match).

10. IDs, match keys, deduplication, lineage

Concern	Mechanism
Internal IDs	UUID vendor_id / job_id
External match	Unique (provider_instance_id, entity_type, external_id)
Inbox dedupe	Unique (provider_instance_id, event_id)
Outbox dedupe	Unique (provider_instance_id, idempotency_key)
Payload lineage	Payload hash + entity version on identity/inbox
Concurrent duplicates	DB uniqueness + claim lease

No fuzzy people/address matching.

11. Inbound sequence and transaction boundaries

1. **Webhook:** verify HMAC over raw bytes → normalize → receipt TX (insert inbox) → 202 (accept/duplicate).
2. **Poll/sync:** provider HTTP **outside** DB TX → synthetic event id → same receipt path.
3. **Process:** claim TX → apply TX (canonical + identity + audit) — **no HTTP** in apply.

Sequences: docs/architecture/inbound-sequence.mmd, ADR-005.

12. Outbound dispatch / outbox / reconciliation

1. Eligibility: Job qualified, Vendor approved, required fields present.
2. Outbox TX inserts pending message with stable idempotency key — **no HTTP**.
3. Worker claims → POST work-order → on ambiguous timeout, GET by client reference (Job ID) **before** any retry with the **same** key.
4. Complete TX: identity link + Job dispatched + outbox completed.

Sequence: docs/architecture/outbound-sequence.mmd.

13. Ordering, idempotency, dependencies, replay, DLQ

Scenario	Behavior
Stale / older version	Audited ignored_stale; no regression
Equal version + same hash	Ignored stale / no mutation
Equal version + different hash	version_payload_conflict (no overwrite)
Unknown contractor	WaitingForDependency; no partial Job
Exhausted attempts/age	DeadLettered + failure history
Replay	Operator-gated; retains event identity; idempotent if already completed

14. Failure, retry, rate limit, circuit, degraded mode

Single HTTP retry owner: FieldFlow resilience pipeline (concurrency → retry → circuit → attempt timeout → transport). Default worst-case per call: 1 + MaxRetryAttempts (4). Workers own durable business attempts separately.

Trigger	HTTP retry	Visible status	Durable disposition
Transient 5xx / timeout	Bounded + jitter	Degraded → Offline if circuit opens	RetryAt → DLQ when exhausted
429 + Retry-After	Honour (capped)	Degraded on trend	RetryAt
Validation 400/422	None	Unchanged	DeadLetter
401/403	None	NeedsAttention	NeedsAttention
Circuit open	Fail fast	Offline	RetryAt / short-circuit
Ambiguous POST	No blind retry	Degraded if storm	Reconcile + same key
Auth / circuit healthy, backlog clear	—	Healthy	—

GET /health/live and /health/ready **ignore** FieldFlow availability.

15. Audit, correlation, monitoring, alerts, health

- Correlation: X-Correlation-Id (validated; invalid replaced).
- Append-only audit_events with sanitized fields only.
- Meter/ActivitySource PRG.Proof360.Integrations — low-cardinality labels only.
- Proposed alerts (rules only): circuit Offline, auth NeedsAttention, DLQ depth, backlog age, sync freshness, 429 ratio, webhook reject spike — see docs/architecture/observability.md.

16. Authentication, secrets, privacy, retention

Prototype	Production evolution
Static API key + HMAC secret via env / user-secrets	OAuth client credentials + vault + rotation
.env.example placeholders only	Managed secret store; least privilege
No PII/secrets in logs/health/audit	Retention policy for inbox payloads TBD with PRG

Never log Authorization, signatures, raw bodies, phone, or email.

17. Approval gates

Gate	Phase 1 rule
Vendor approved	Explicit Proof360 decision only
Outbound dispatch	qualified Job + approved Vendor
Payments / funding	Blocked
Terminal reopen	Not implemented
DLQ replay	Operator-triggered; idempotent

18. Second-provider evolution

1. Add `PRG.Proof360.Integrations.<Provider>` implementing Core capability ports.
2. Register in API composition; keep Domain/Application unchanged.
3. Scope identity/inbox uniqueness by `provider_instance_id`.
4. Reuse disposition, outbox, health, and audit patterns.

19. Acceptance criteria (observable)

#	Criterion	Evidence
1	Canonical Vendor/Job/Transcript field contracts match assignment	CanonicalFieldContractTests, CanonicalColumnTests
2	Domain/Application have no FieldFlow DTO dependency	FieldFlowDtoBoundaryTests, ProjectDependencyTests
3	Repeat contractor poll → one Vendor + identity link	Repeating_contractor_snapshot_yields_one_vendor_and_link
4	Unknown contractor waits; no partial Job; exhaustion → DLQ	Unknown_contractor_waits_*, Exhausted_unknown_contractor_*
5	Valid webhook durable before ack; invalid HMAC mutates nothing	WebhookSecurityTests
6	≥10 concurrent duplicate webhooks → one inbox + one Job	Concurrent_duplicate_webhooks_create_one_inbox_and_one_job
7	Newer status applies; older after terminal cannot regress	Newer_status_then_terminal_blocks_older_event_regression
8	Equal-version same payload no mutation; different → conflict	Equal_version_*
9	Outbox dispatch once; ambiguous POST reconciles without duplicate	OutboundDispatchTests
10	Retry-After / circuit / liveness independent of FieldFlow	HttpResiliencePipelineTests, HostHealthEndpointTests
11	DLQ replay idempotent; audit omits secrets/PII markers	AuditReplayObservabilityTests
12	Local mock + API sync + connector health without credentials	Prompt 10 smoke / this README demo

20. Tradeoffs and production evolution

Tradeoff	Phase 1	Production
SQLite + in-process workers	Fast local proof	PostgreSQL + hosted workers / broker
Static secrets	Demo safety	Vault + rotation
In-memory circuit state	Process-scoped	Shared/distributed breaker if multi-instance
Admin replay token gate	Local ops	Real authz + change control
Accounting deferred in code	Field service proves patterns	Next commercial bet (ADR-008)

Residual risks: assumed FieldFlow schemas; single-instance workers; no load test; alert routing not wired.

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